Convolutional Neural Networks

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Today’s Class

- What is convolution?
  - Brief remark about traditional approaches (transforms)
- Fei Fei Li’s ImageNet Challenge

- Handcrafted classifiers → Convolutional Networks
  - Remarks about ConvNet structure

- Multiclass classification
- Classification metrics

What is "Convolution?"

- Given a function $f(t)$ and a "filter" $\mathrm{w}(t)$, the *convolution* of the two functions, $\mathrm{g} = f * h$, is another function given by

$$g(t) = \int_{-\infty}^{\infty} f(x)w(t-x)dx$$

- In discrete time,

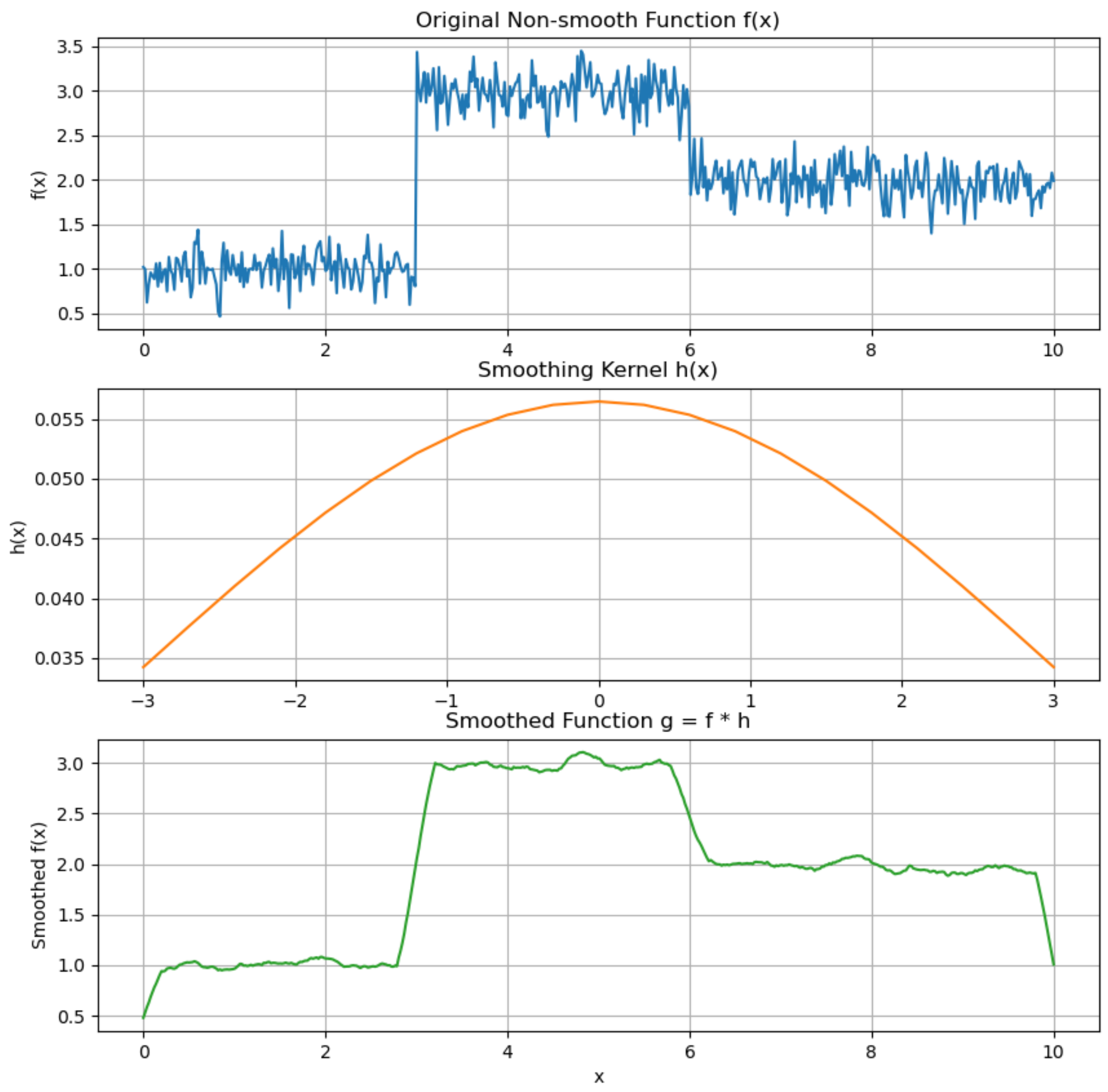
$$g(t) = \sum_{x=-\infty}^{\infty} f(x)w(t-x)$$

What is "Convolution?"

- In discrete time,

$$g(t) = \sum_{x=-\infty}^{\infty} f(x)w(t-x)$$

- Can be used for many filtering operations
  - smoothing,
  - removing high-frequencies
  - removing low-frequencies
  - preserving a certain "band" of frequencies



What is “Convolution?”

- For images, over SPACE

$$g(i,j) = \sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} f(x,y)w(x-i,y-j)$$

- Assume grayscale images
  - $f(i,j)$: intensity of pixel (i,j) in input image f
  - $g(i,j)$: intensity of pixel (i,j) in output image g
  - $w(x,y)$: value of $(i,j)^{th}$ element of filter

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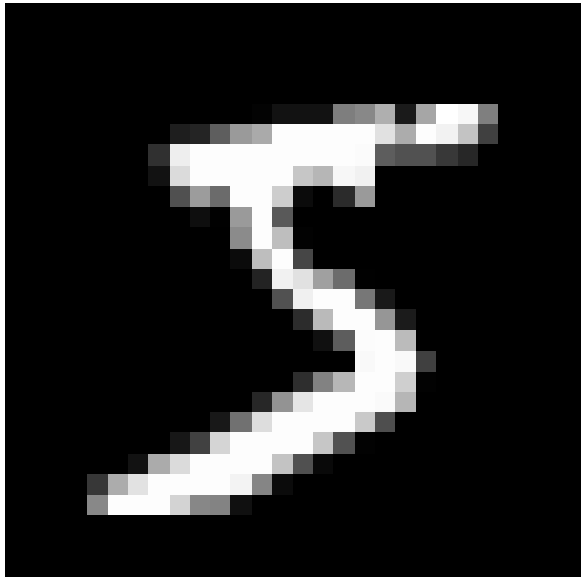
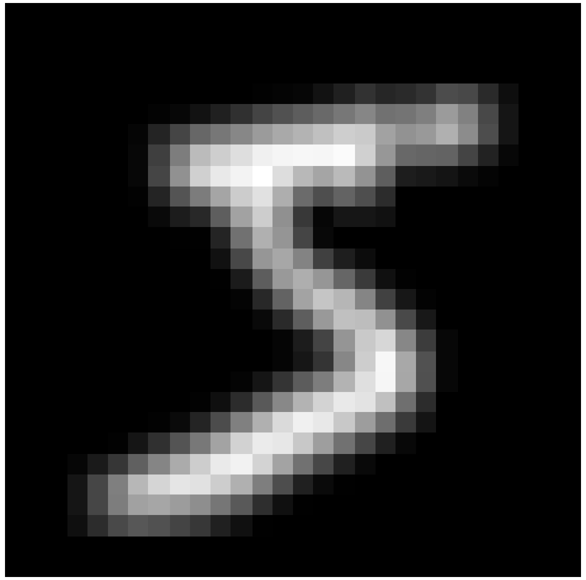
Original MNIST Digit



Smoothed (3×3 Box Filter)

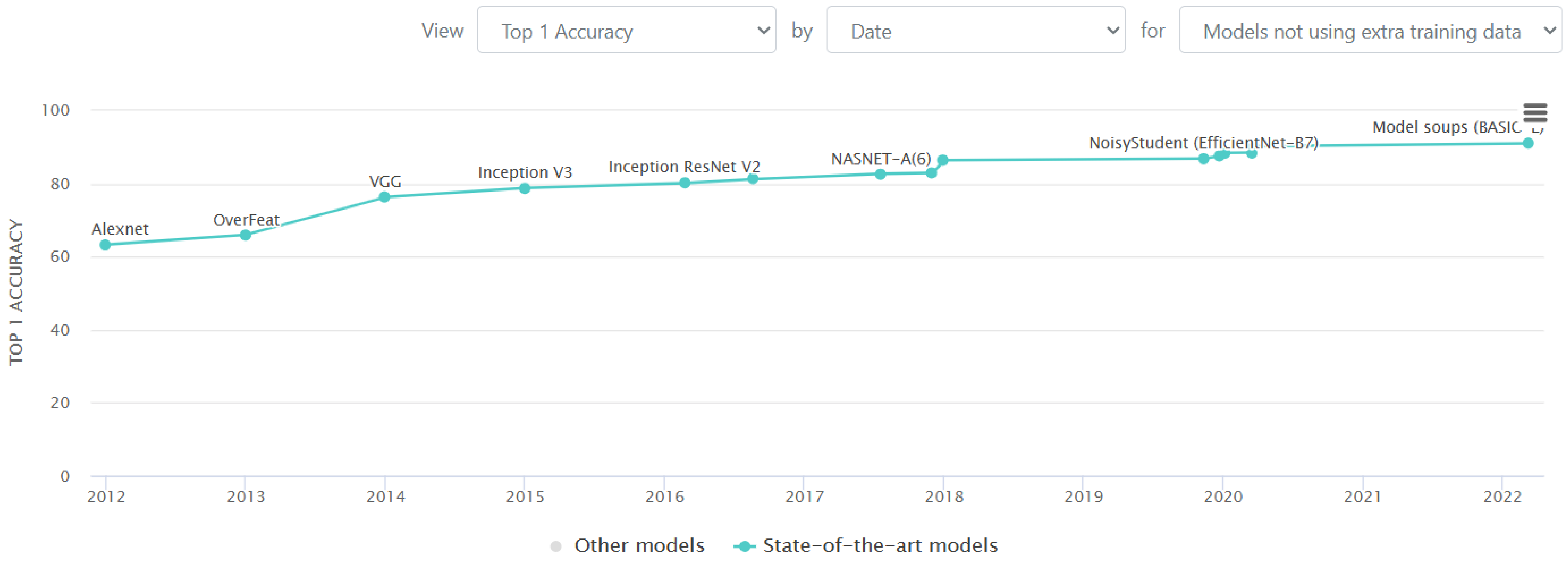


The ImageNet Challenge

- Major benchmark dataset for Computer Vision
- Compiled by Prof. Fei Fei Li
- Challenge: 2010 – 2017
  - 1.4 Million images
  - 1000 classes
  - Labeling using Amazon Mechanical Turk
  - Human accuracy $74\% - 89\%$

  (*not* conducted under CRT)

The ImageNet Challenge



Source: PapersWithCode

Mammalian Vision

- CNNs inspired by the mammalian visual cortex
- Local Receptive Field
  - D.H. Hubel and T. Wiesel (1959)
  - Overlap and tile

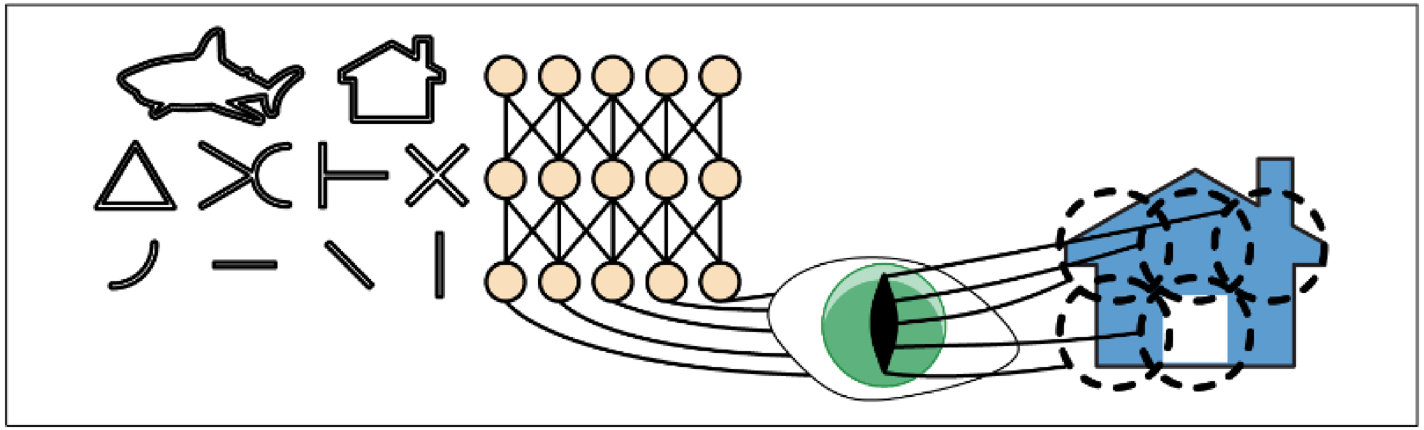
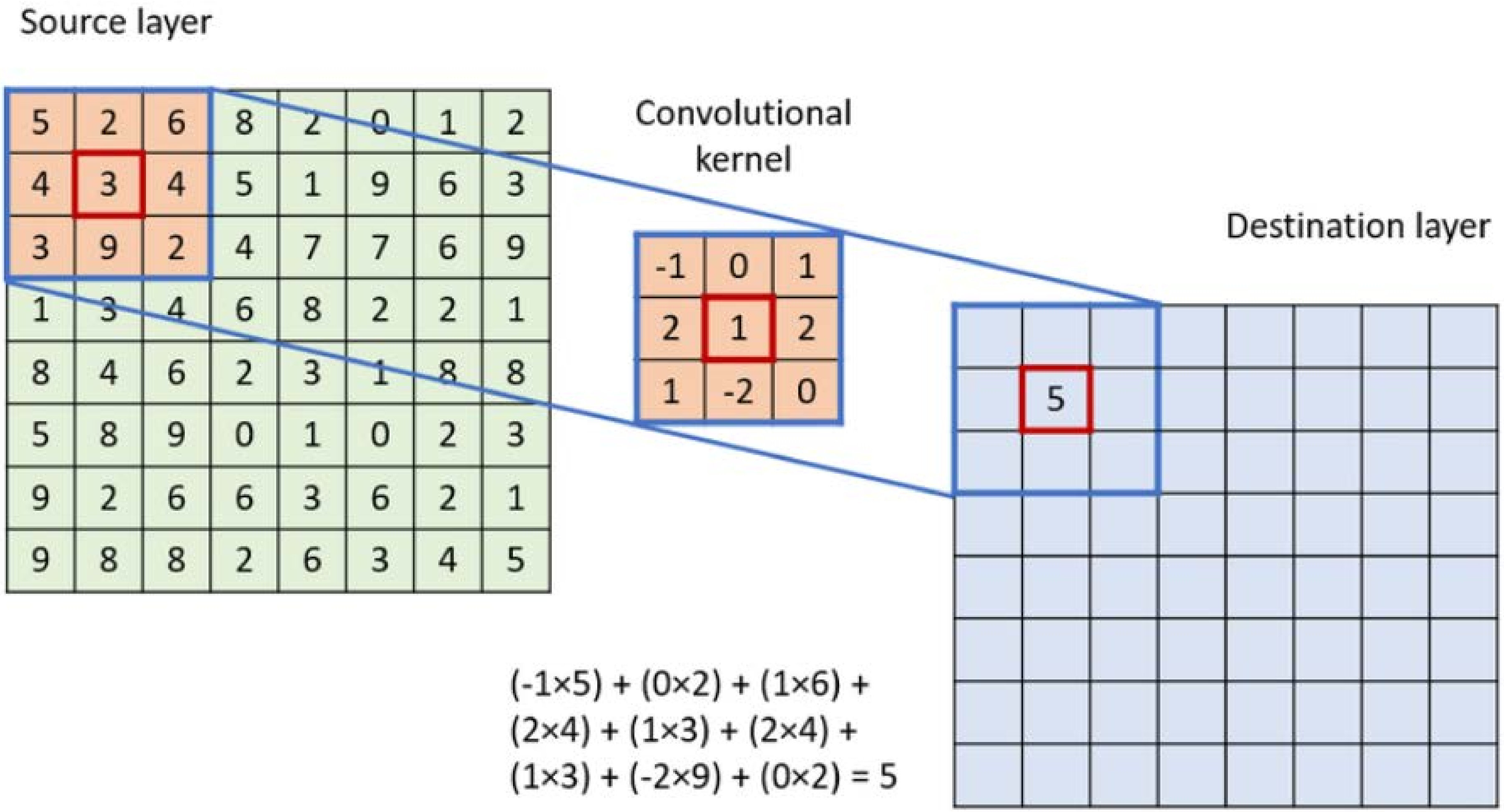


Figure 14-1. Biological neurons in the visual cortex respond to specific patterns in small regions of the visual field called receptive fields; as the visual signal makes its way through consecutive brain modules, neurons respond to more complex patterns in larger receptive fields

- "Complex cells"
  - larger receptive fields
  - react to more complex patterns: combinations of lower-level patterns
  - higher neurons connected only to nearby lower-level neurons

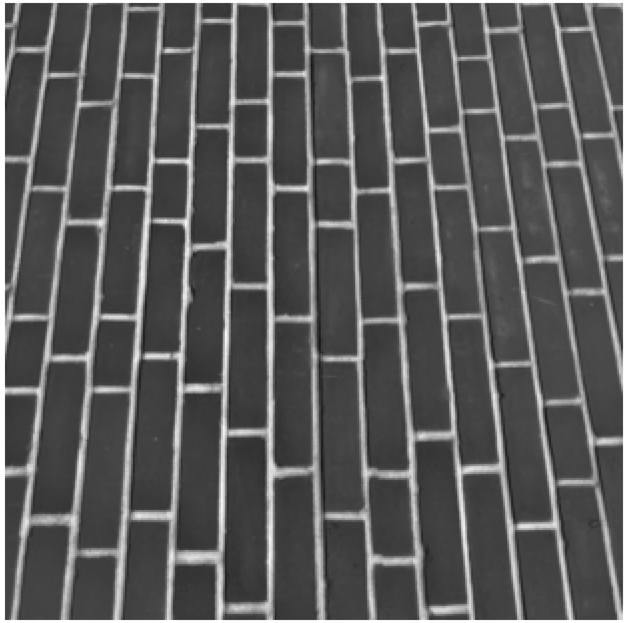
- Yann LeCun et al (LeNet, 1998)
  - The convolutional layer.

Convolutional Layers



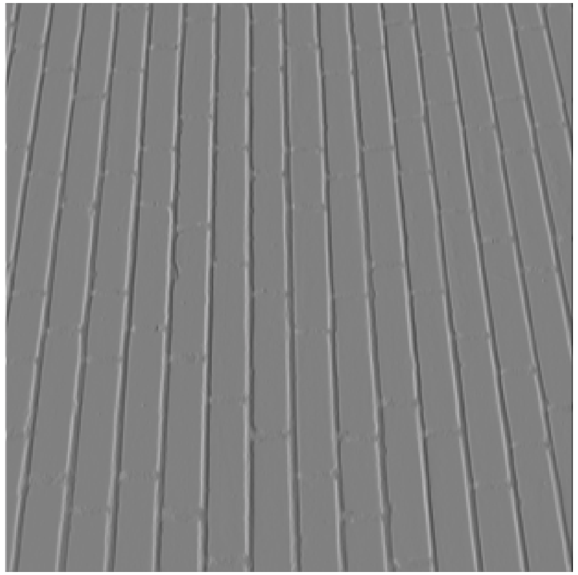
Examples

Original Image



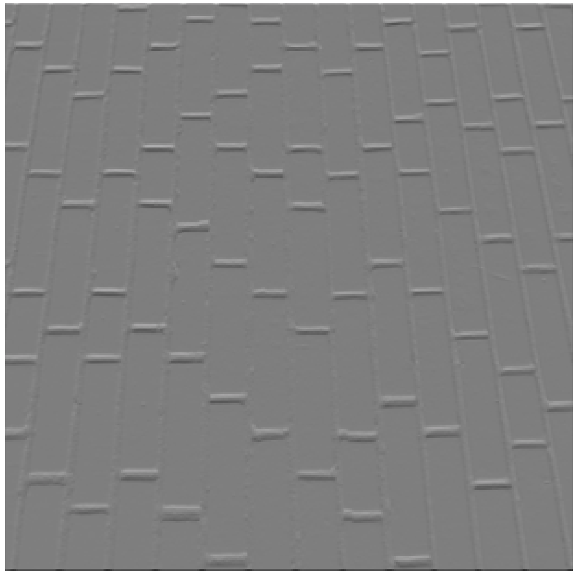
$$\begin{matrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{matrix}$$

Vertical Edges



$$\begin{matrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{matrix}$$

Horizontal Edges



Convolutional Layers

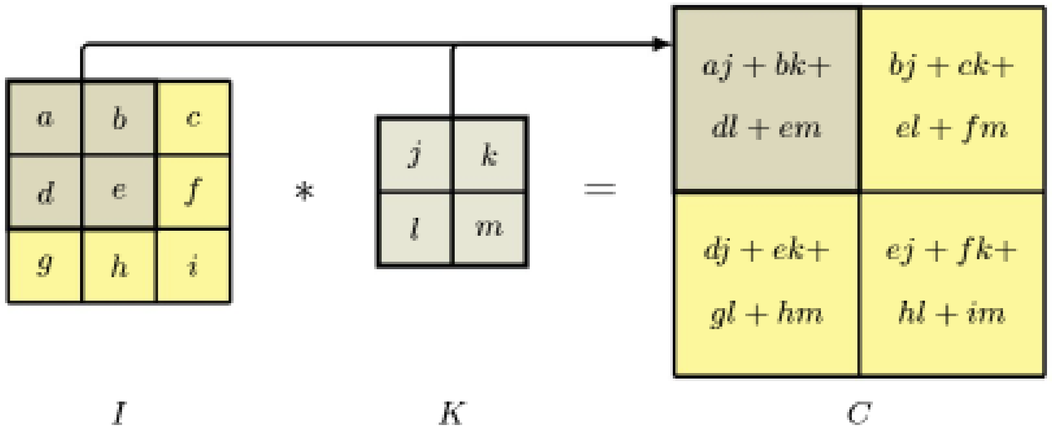


Figure 10.3 Example of a 3×3 image convolved with a 2×2 filter to give a resulting 2×2 feature map.

- Output of a Conv layer

$$Y(i,j) = \sum_{m} \sum_{l} X(i+l, j+m) W(l,m)$$

  - $X(i,j)$: intensity of $(i,j)^{th}$ pixel in input
  - $W(l,m)$: $(l,m)^{th}$ weight in Conv layer
  - $Y(i,j)$: intensity of $(i,j)^{th}$ pixel in input

- Nonlinearity $h(\cdot)$
  - ReLu activation

Max Pooling

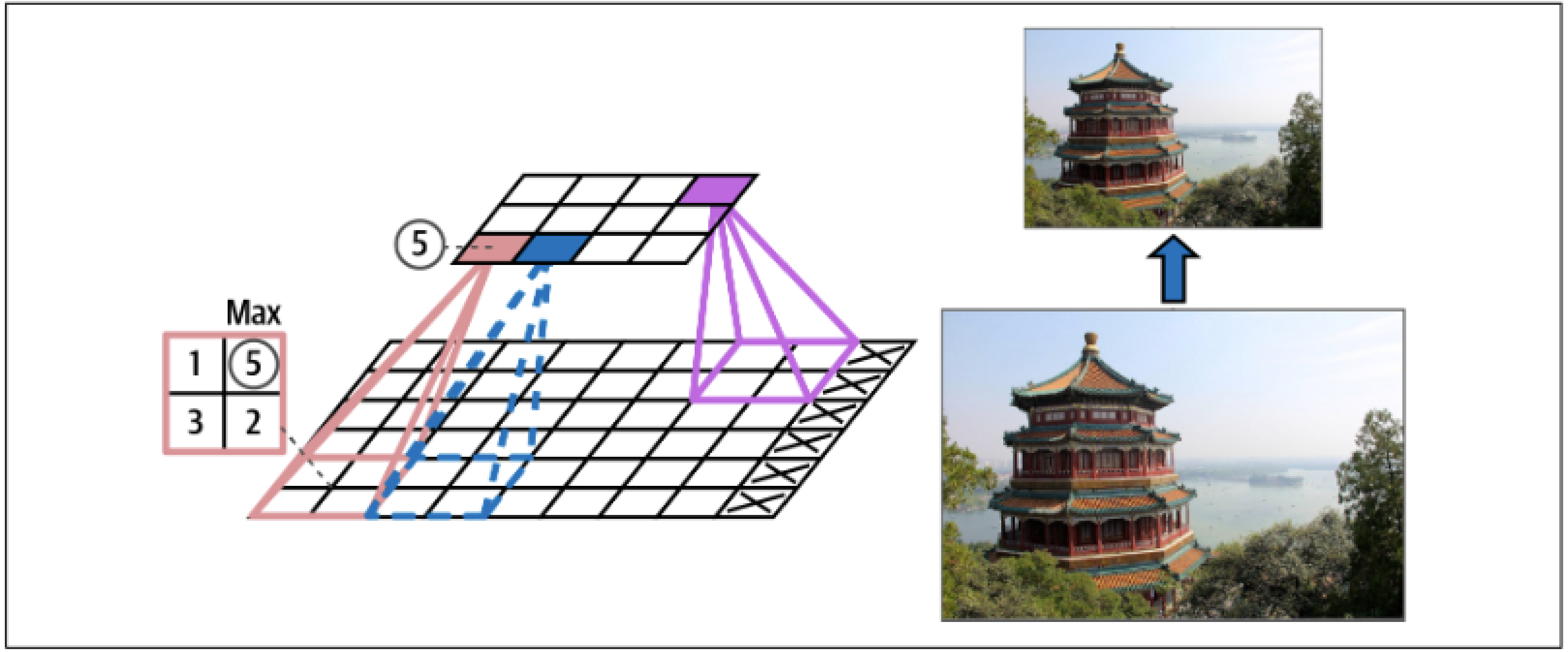
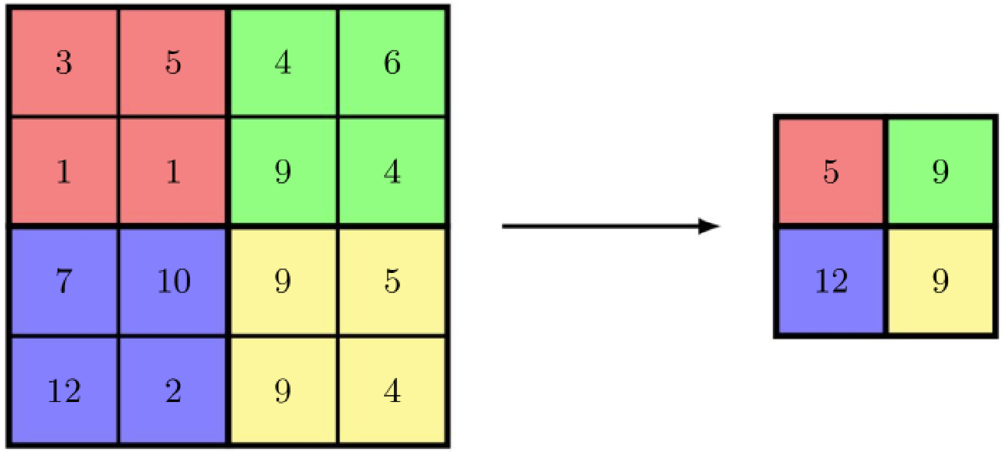
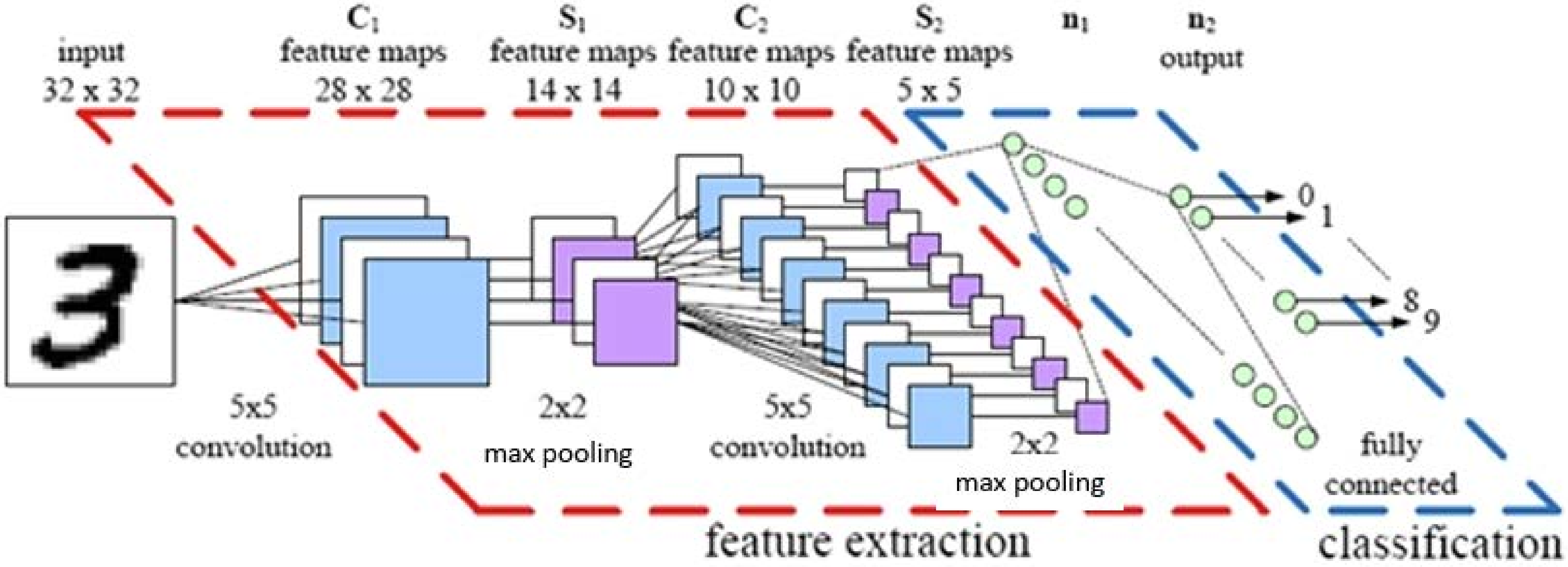


Figure 14-9. Max pooling layer (2 × 2 pooling kernel, stride 2, no padding)

- Pooling layers “*subsample*” (shrink)
- Goals
  - local equivariance
  - reducing computational load, memory usage,
  - reducing number of parameters ⇒ risk of overiftting



CNN Architecture



Benefits of this Architecture

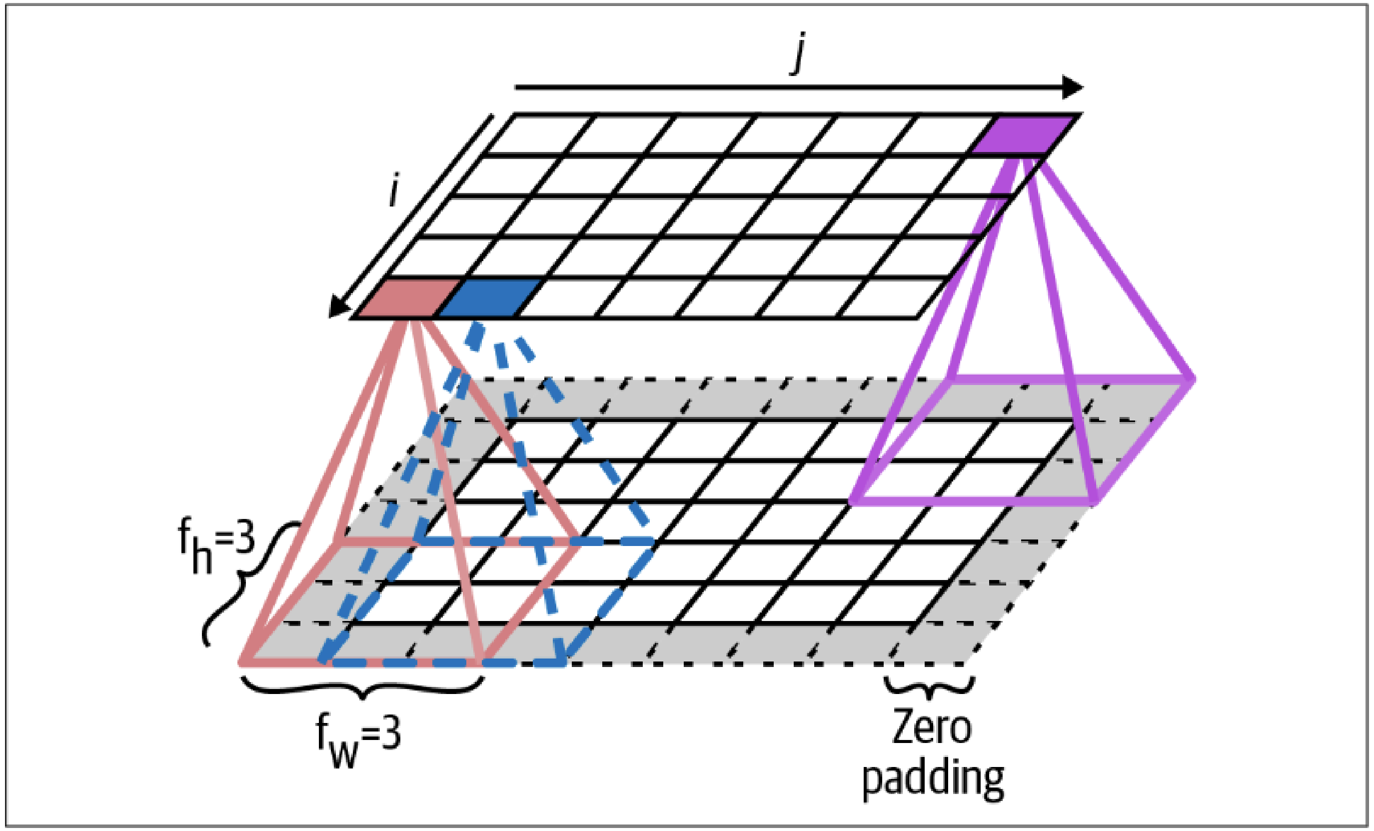
- Sparse interactions
  - Fewer connections
- Parameter sharing
  - Fewer weights to train
  - Backprop works
- Locally equivariant representation
  - Invariant to translations

Zero padding

- Prevents rapid shrinkage
- Size of output = Size of input

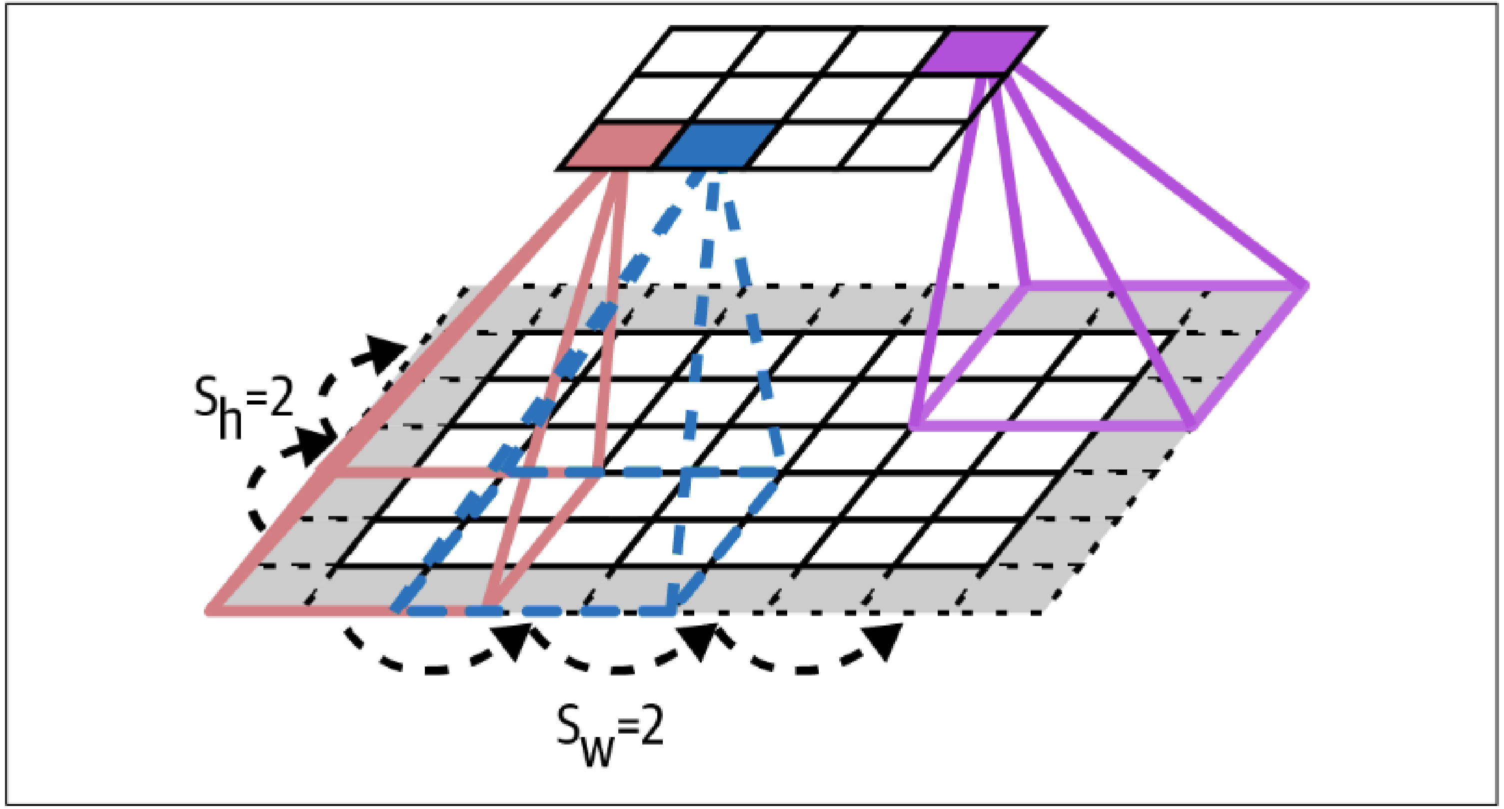
- Tensorflow
  - `padding = "valid"`
  - `padding = "same"`

- $O = I + 2p - h + 1$



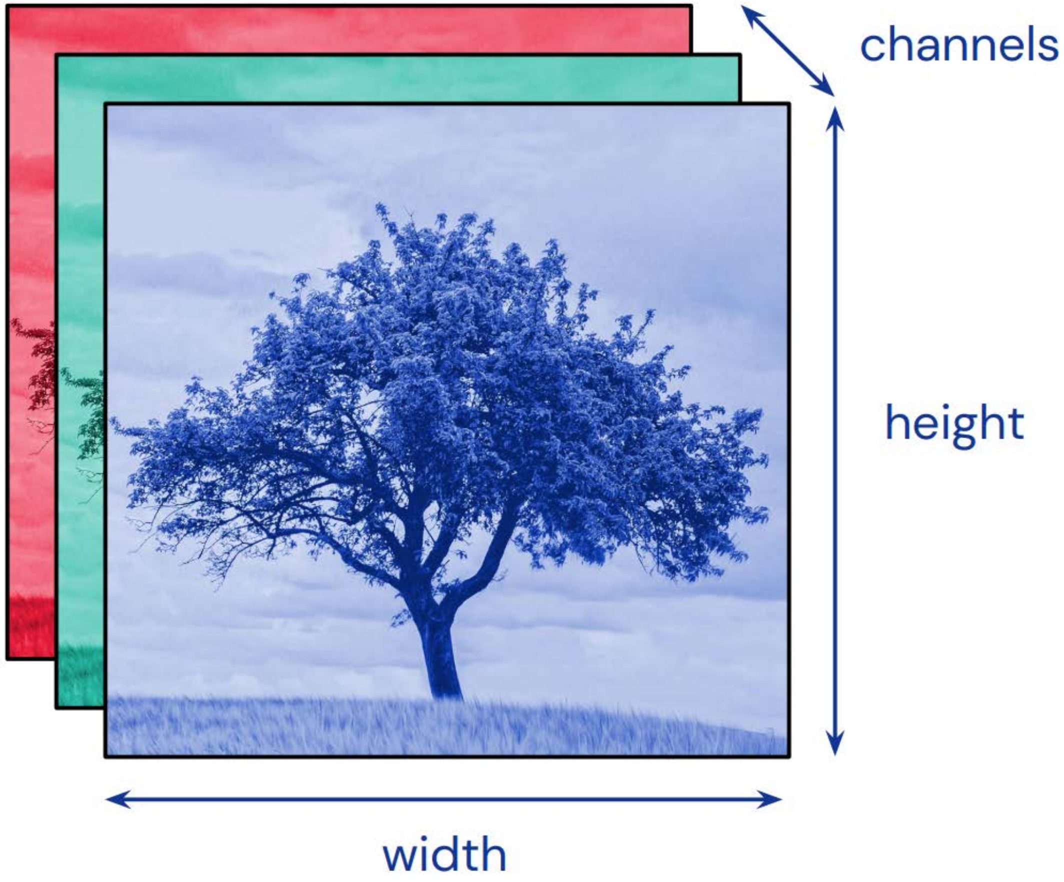
Strided Convolution

- Images much larger than kernels
  - Slide with larger step sizes.
- $O = \left\lfloor \frac{I+2p-h}{s} \right\rfloor + 1$
  - $\lfloor x \rfloor$: “floor”
  - Output dimensions are *roughly* $\frac{1}{s}$ *times* the input dimensions.



Examples

Color Images



Source: Sander Dieleman, DeepMind

Vectors to “Tensors”